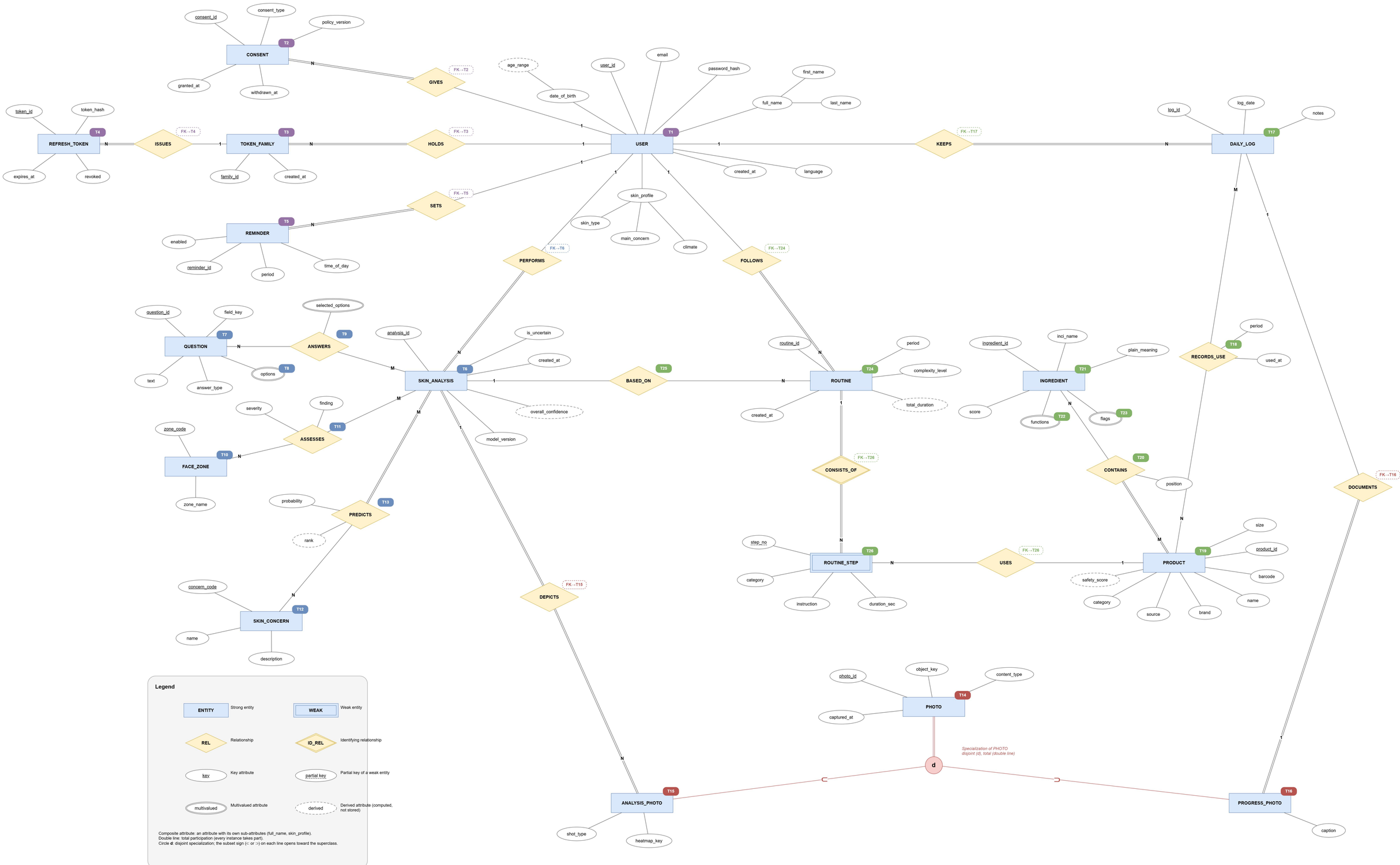
From the EER model (top) to the relational schema (bottom)

Every EER element that becomes a table carries a coloured badge **Tn**; the same badge marks the table it produced. A relationship that becomes a foreign key carries a dashed badge **FK→Tn** pointing to the table that holds the key. The mapping table (bottom right) gives the rule used for each element. Colours: purple account and security, blue skin analysis, red photos, green products, routines and log.

AI-Assisted Skincare Companion App: Enhanced Entity-Relationship (EER) model

Chen notation. Cardinality ratios 1, N, M; a double line marks total participation.



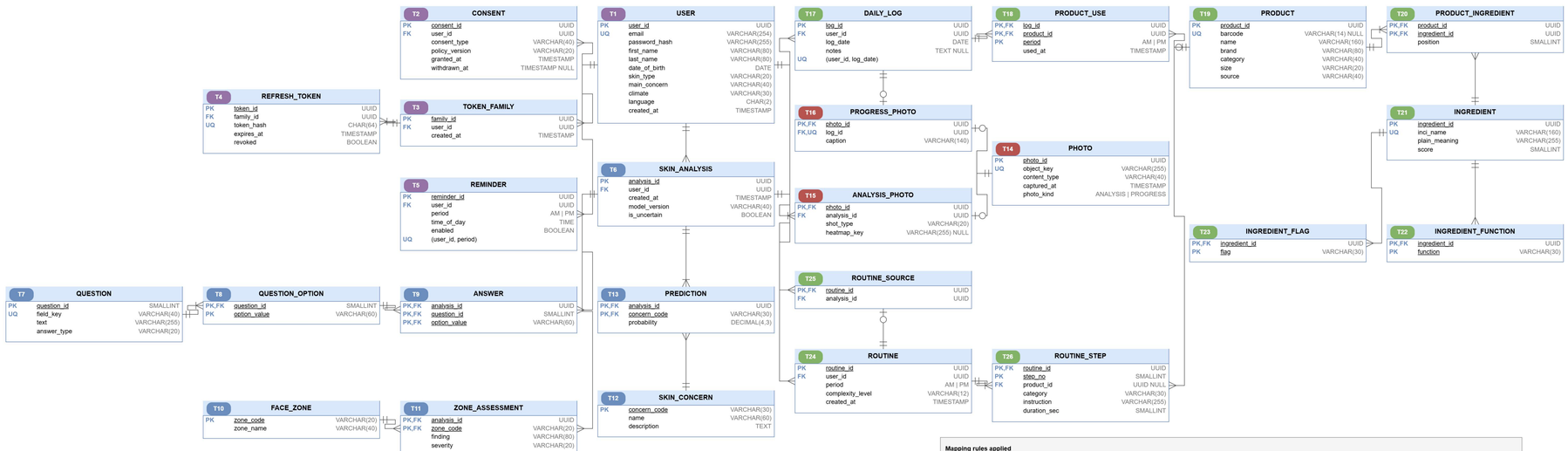
Legend

ENTITY	Strong entity	WEAK	Weak entity
REL	Relationship	ID_REL	Identifying relationship
key	Key attribute	partial key	Partial key of a weak entity
multivalued	Multivalued attribute	derived	Derived attribute (computed, not stored)

Composite attribute: an attribute with its own sub-attributes (full_name, skin_profile).
Double line: total participation (every instance takes part).
Circle **d**: disjoint specialization; the subset sign (⊂ or ⊃) on each line opens toward the superclass.

Relational schema mapped from the EER model, normalized to BCNF

PK primary key (underlined), FK foreign key, UQ unique. Crow's-foot ends: bar = exactly one, circle = optional, fork = many.



Mapping rules applied
1. Each strong entity becomes a table; its key becomes the primary key.
2. Composite attributes (full_name, skin_profile) are flattened into simple columns.
3. Derived attributes (age_range, total_duration, safety_score, overall_confidence, rank) are computed, not stored.
4. 1:N relationships: foreign key on the N side (e.g. SKIN_ANALYSIS.user_id).
5. M:N relationships become junction tables that carry the relationship attributes (ZONE_ASSESSMENT, PREDICTION, PRODUCT_USE, PRODUCT_INGREDIENT); ANSWER holds one row per selected option (multivalued relationship attribute), so a skipped question has no row.
6. Multivalued attributes become their own tables (QUESTION_OPTION, INGREDIENT_FUNCTION, INGREDIENT_FLAG).
7. Weak entity ROUTINE_STEP: primary key = owner key routine_id + partial key step_no.
8. Specialization of PHOTO (disjoint, total): one table per class; each subclass reuses the superclass key (photo_id is both PK and FK) and photo_kind records the subclass.
9. 1:1 DOCUMENTS: unique foreign key PROGRESS_PHOTO.log_id.
10. Optional 1:N BASED_ON: own table ROUTINE_SOURCE (no NULL; no dependency analysis_id → user_id inside ROUTINE).
All 26 tables are in Boyce-Codd normal form (BCNF).

Mapping table

Badge	EER element	Construct	Rule	Relational table / column
T1	USER	strong entity	R1, R2, R3	**USER**
T2	CONSENT	strong entity	R1	**CONSENT**
T3	TOKEN_FAMILY	strong entity	R1	**TOKEN_FAMILY**
T4	REFRESH_TOKEN	strong entity	R1	**REFRESH_TOKEN**
T5	REMINDER	strong entity	R1	**REMINDER**
T6	SKIN_ANALYSIS	strong entity	R1, R3	**SKIN_ANALYSIS**
T7	QUESTION	strong entity	R1	**QUESTION**
T8	QUESTION.options	multivalued attribute	R6	**QUESTION_OPTION**
T9	ANSWERS (M:N) + selected_options	relationship, multivalued attribute	R5, R6	**ANSWER**
T10	FACE_ZONE	strong entity	R1	**FACE_ZONE**
T11	ASSESSES (M:N)	relationship	R5	**ZONE_ASSESSMENT**
T12	SKIN_CONCERN	strong entity	R1	**SKIN_CONCERN**
T13	PREDICTS (M:N)	relationship	R5, R3	**PREDICTION**
T14	PHOTO	superclass	R8	**PHOTO**
T15	ANALYSIS_PHOTO	subclass	R8	**ANALYSIS_PHOTO**
T16	PROGRESS_PHOTO	subclass	R8	**PROGRESS_PHOTO**
T17	DAILY_LOG	strong entity	R1	**DAILY_LOG**
T18	RECORDS_USE (M:N)	relationship	R5	**PRODUCT_USE**
T19	PRODUCT	strong entity	R1, R3	**PRODUCT**
T20	CONTAINS (M:N)	relationship	R5	**PRODUCT_INGREDIENT**
T21	INGREDIENT	strong entity	R1	**INGREDIENT**
T22	INGREDIENT.functions	multivalued attribute	R6	**INGREDIENT_FUNCTION**
T23	INGREDIENT.flags	multivalued attribute	R6	**INGREDIENT_FLAG**
T24	ROUTINE	strong entity	R1, R3	**ROUTINE**
T25	BASED_ON (1:N, optional)	relationship	R10	**ROUTINE_SOURCE**
T26	ROUTINE_STEP	weak entity	R7	**ROUTINE_STEP**
FK→T2	GIVES (1:N)	relationship	R4	CONSENT.user_id
FK→T3	HOLDS (1:N)	relationship	R4	TOKEN_FAMILY.user_id
FK→T4	ISSUES (1:N)	relationship	R4	REFRESH_TOKEN.family_id
FK→T5	SETS (1:N)	relationship	R4	REMINDER.user_id
FK→T6	PERFORMS (1:N)	relationship	R4	SKIN_ANALYSIS.user_id
FK→T15	DEPICTS (1:N)	relationship	R4	ANALYSIS_PHOTO.analysis_id
FK→T16	DOCUMENTS (1:1)	relationship	R9	PROGRESS_PHOTO.log_id (UQ)
FK→T17	KEEPS (1:N)	relationship	R4	DAILY_LOG.user_id
FK→T24	FOLLOWS (1:N)	relationship	R4	ROUTINE.user_id
FK→T26	CONSISTS_OF (identifying)	relationship	R7	ROUTINE_STEP.routine_id (part of PK)
FK→T26	USES (N:1)	relationship	R4	ROUTINE_STEP.product_id

Rules (same numbering as the note "Mapping rules applied" of the relational schema)
R1 entity to table, key to primary key
R2 composite attribute flattened
R3 derived attribute not stored
R4 1:N relationship: foreign key on the N side
R5 M:N relationship: junction table with its attributes
R6 multivalued attribute: own table
R7 weak entity: owner key + partial key
R8 specialization: one table per class, shared key
R9 1:1 relationship: unique foreign key
R10 optional 1:N: own table (no NULL, BCNF)